A Mini Project Report on

Telecom Attrition Risk Prediction

Using Machine Learning

submitted to

PYTHON PROGRAMMING
(CSE236P)
by

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SCHOOL OF ENGINEERING AND TECHNOLOGY
CHRIST (DEEMED TO BE UNIVERSITY)

Academic Year: 2025-2026

CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

School of Engineering and Technology
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CERTIFICATE

This is to certify that ALEN JOJO, AYUSH KUMAR, JOSBY JAMES, bearing Register Nos. 2562005, 2562015, 2562032, have satisfactorily completed the project work in **CSE236P - PYTHON PROGRAMMING** prescribed by the **Department of AI and Data Science Engineering**, CHRIST (Deemed to be University) for the II semester B.Tech course during the academic year **2025-26**.

Signature of Faculty In-charge **Signature of the HOD**

Date: 26/02/2026

Place: **Bangalore**

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Vision

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Core Values

Faith in God
Moral Uprightness
Love of Fellow Beings
Social Responsibility
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Department Vision

"To excel in human-centered AI and data-driven innovation"

Department Mission

M1. *Empowering individuals to ethically harness data and AI through accessible and value-driven curriculum.*

M2. *Foster a dynamic research environment that advances innovative and impactful solutions for the betterment of global well-being.*

M3. *Innovate scientific knowledge and entrepreneurship through academia and Industry collaborations.*

Program Educational Objectives (PEOs)

PEO1 - Professional Acumen: Understand, analyze and design solutions with professional competency for the real-world problems

PEO2 - Critical Analysis: Develop software/embedded solutions for the requirements, based on critical analysis and research.

PEO3 - Team work: Function effectively in a team and as an individual in a multidisciplinary/multicultural environment.

PEO4: Life Long Learning: Accomplish holistic development comprehending professional responsibilities

Program Outcomes (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1. Artificial Intelligence: To utilize artificial intelligence principles to solve real world problems

PSO2. Machine Learning To select suitable machine learning algorithms for the given problem

PSO3. Service Learning Analyze Social Relevant Problems and design solutions through Service Learning

Evaluation Rubrics

• Planning & Time Management	10 Marks
• Time and Effort & Team Work	10 Marks
• Evidence of Learning and Risk Factor	10 Marks
• Degree of Difficulty	10 Marks
• Report Preparation	10 Marks

Contents

Chapter No	Chapter Title	Page No
1.	**Introduction**	**8**
1.1	Problem Statement	8
1.2	Objectives	8
1.3	Literature Review	9
2.	**Execution**	**10**
2.1.	Key Concepts Used	10
2.2.	Modules and Description	11
2.3.	Program Source Code	12
3.	**Outcomes**	**16**
3.1.	Result and Output	16
3.2.	Conclusions	21
3.3.	Scope for Future Work	22
4.	**References**	**23**

CHAPTER 1

Introduction

PROBLEM STATEMENT

In the telecommunication market, which is highly competitive, the cost of acquiring a new customer is much higher than the cost of retaining an existing customer. Customer Churn, or the rate at which customers leave a service provider, has a direct impact on revenue and market share.

The company does not have an automated system in place to identify customers who are "at risk". Without a predictive system, the marketing department cannot reach out to customers with loyalty programs or special promotions before they choose to go with a competitor. There is a great need to use customer data(demographic, account, and usage information) to predict the likelihood of customer churn and to understand the factors that influence customer churn.

OBJECTIVES

The primary goal of this analysis is to build a robust machine learning pipeline to predict and understand customer behavior. The specific objectives are:

A. Data Cleaning & Feature Engineering

- Handle Data Integrity: Identify and resolve missing values (specifically in TotalCharges) and remove redundant features like customerID.
- Data Transformation: Encode categorical variables into a machine-readable format using techniques like One-Hot Encoding and binary mapping.
- Feature Scaling: Standardize numerical features to ensure distance-based models (like KNN and SVM) perform optimally.

B. Exploratory Data Analysis (EDA)

- Identify Patterns: Visualize the relationship between tenure, MonthlyCharges, and Churn to find behavioral trends.
- Correlation Analysis: Use heatmaps to determine which numerical factors have the strongest linear relationship with the decision to leave.

C. Model Development & Comparison

- Algorithm Benchmarking: Train and compare multiple classification algorithms including Logistic Regression, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Random Forest.
- Hyperparameter Tuning: Utilize GridSearchCV to optimize model parameters for the best performance.
- Class Imbalance Management: Implement stratify during data splitting and use class_weight='balanced' in Random Forest to account for the fact that "Churn" cases are typically less frequent than "Non-Churn" cases.

D. Model Evaluation & Insights

- Performance Metrics: Evaluate models not just on Accuracy, but also on Precision, Recall, F1-Score, and ROC-AUC to ensure the model effectively identifies churners (True Positives) without excessive False Alarms.
- Visual Validation: Generate Confusion Matrices and ROC Curves to visualize the trade-off between sensitivity and specificity.

LITERATURE REVIEW

Predicting customer churn is a highly important task in the telecommunication industry, where market maturity makes customer retention far more economical than customer acquisition. This review aims to assess the current status of predictive analysis, from traditional statistical approaches to machine learning and the need for proper data preprocessing in high-dimensional data.

The underlying rationale for customer churn prediction is based on the "Customer Lifetime Value" hypothesis. Studies have indicated that even a slight improvement in customer retention can lead to a substantial boost in profitability. In the telecommunication industry, intense competition requires the use of automated systems that can detect behavioral "warning signs" of churn, such as escalating monthly bills or a reduction in tenure, before the customer actually leaves the service.

The literature groups churn prediction models into three main families:

- Baseline Statistical Models: Logistic Regression is a classic choice because of its interpretability and ability to perform well on binary classification problems.
- Ensemble and Tree-Based Models: Random Forest [1] is frequently mentioned as the "gold standard" model for structured data such as telecom usage. It is robust to non-linear relationships and inherently robust to outliers and missing data.
- Distance-Based Models: K-Nearest Neighbors (KNN) and Support Vector Machines (SVM) [2] are robust models but extremely sensitive to the scale of features. Consensus among scholarly literature emphasizes the need for standardization (using tools such as StandardScaler [3]) to avoid numerical biases.

One of the common observations in telecom literature is the issue of "dirty data." In particular, the TotalCharges column is known to have missing or incorrectly formatted data. Modern literature recommends the use of median imputation as a robust method to impute missing data without affecting the distribution [4]. Additionally, class imbalance is addressed using stratified splitting and class weights to favor high recall of the minority class [5]. The use of ROC-AUC analysis [6] is preferred over simple accuracy to evaluate how well a model distinguishes between churners and non-churners.

CHAPTER 2

Execution

KEY CONCEPTS USED

Data Preprocessing & Cleaning

- **Type Casting:** The process of changing data types from one form (such as text) to another (such as numbers) to enable mathematical calculations.
- **Median Imputation:** The use of the middle value of a set of data to replace missing values in order to avoid gaps in the analysis without affecting the outcome.
- **Standardization:** The process of adjusting numerical values to a standard scale to avoid features with high numbers from dominating the decision-making process.

Feature Engineering

- **One-Hot Encoding:** Converting categorical text labels into binary 0s and 1s so machine learning algorithms can process non-numeric information.
- **Dummy Variable Trap Prevention:** Removing one redundant column after encoding to prevent mathematical errors caused by perfectly correlated variables.

Model Training & Selection

- **Ensemble Learning:** Combining the predictions of multiple individual models (like many decision trees) to create one more accurate and stable "forest."
- **Grid Search:** An automated process that tests various settings for a model to find the most effective configuration for a specific dataset.
- **Stratification:** Ensuring that your training and testing sets maintain the same ratio of "churners" to "non-churners" as the original data.
- **Class Weighting:** Assigning higher importance to the minority class (churners) to ensure the model doesn't ignore them in favor of the majority.

Evaluation Metrics

- **Precision:** The percentage of predicted churners who were actually churners, measuring the model's "exactness."
- **Recall:** The percentage of all actual churners that the model successfully identified, measuring its "completeness."
- **F1-Score:** A single balanced metric that combines both precision and recall to evaluate overall performance.

- **ROC-AUC:** A score representing the model's ability to distinguish between two different groups, where 1.0 is perfect and 0.5 is random.
- **CV ROC-AUC:** The average of multiple ROC-AUC scores calculated across different data subsets to ensure the model's predictive power is consistent and not biased by a single random split.

MODULES AND DESCRIPTION

Data Manipulation & Mathematics

- **Pandas (pd):** This is the main library for data structures, and it enables you to import the CSV and to process missing data, deal with the "TotalCharges" column, and perform One-Hot Encoding on the categorical variables.
- **NumPy (np):** This is a core library for scientific computing, and it is the one that enables the high-performance multidimensional array objects used by Scikit-learn to actually carry out the mathematical computations for your models.

Data Visualization

- **Matplotlib (plt):** This is the basic plotting library that is used to create static, interactive, and animated plots such as the ROC curve and bar plots.
- **Seaborn (sns):** This is a high-level plotting library that is built on top of Matplotlib and is used to create attractive statistical graphics such as heatmaps and box plots.

Machine Learning (Scikit-learn)

- **sklearn.model_selection:** Contains tools like train_test_split for dividing data and GridSearchCV or cross_val_score for automated model tuning and robust cross-validation.
- **sklearn.preprocessing:** Features the StandardScaler, which is used to normalize numerical data so all variables (like tenure and charges) are on the same mathematical scale.
- **sklearn.linear_model:** Provides the LogisticRegression algorithm, which serves as a baseline model for binary classification.
- **sklearn.tree:** Implements the DecisionTreeClassifier, a model that makes predictions by splitting data into branches based on feature importance.
- **sklearn.ensemble:** Contains the RandomForestClassifier, GradientBoostingClassifier, and ExtraTreesClassifier, which combine multiple trees to improve prediction accuracy.
- **sklearn.svm:** Implements SVC (Support Vector Classification) to find the optimal boundary or "hyperplane" between churners and non-churners.
- **sklearn.neighbors:** Provides the KNeighborsClassifier, which classifies a customer based on the behavior patterns of their mathematically "nearest" peers.
- **sklearn.metrics:** The evaluation toolbox used to calculate accuracy_score, precision_score, recall_score, f1_score, and roc_auc_score, as well as generating the confusion_matrix and classification_report.

PROGRAM SOURCE CODE

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, cross_val_score, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier, ExtraTreesClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score
from sklearn.metrics import classification_report, confusion_matrix

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

df.head()

print("Dataset Shape:", df.shape)
df.info()
df.isnull().sum()

# Drop customerID (not useful for prediction)
df.drop("customerID", axis=1, inplace=True)

clean_data = df.copy()

# Convert TotalCharges to numeric
clean_data["TotalCharges"] = pd.to_numeric(clean_data["TotalCharges"], errors="coerce")

# Fill missing values using median
clean_data["TotalCharges"].fillna(clean_data["TotalCharges"].median(), inplace=True)

# Convert target variable to numeric
clean_data["Churn"] = clean_data["Churn"].map({"No": 0, "Yes": 1})

clean_data.isnull().sum()

plt.figure(figsize=(6,4))
sns.countplot(x="Churn", data=clean_data)
plt.title("Churn Distribution (0 = No, 1 = Yes)")
plt.show()

plt.figure(figsize=(6,4))
sns.boxplot(x="Churn", y="tenure", data=clean_data)
plt.title("Tenure vs Churn")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.boxplot(x="Churn", y="MonthlyCharges", data=clean_data)
plt.title("Monthly Charges vs Churn")
plt.show()

sns.pairplot(
    clean_data[["tenure", "MonthlyCharges", "TotalCharges", "Churn"]],
    hue="Churn"
)
plt.show()

plt.figure(figsize=(8,6))

numeric_cols = ["SeniorCitizen", "tenure", "MonthlyCharges", "TotalCharges", "Churn"]

sns.heatmap(clean_data[numeric_cols].corr(),
            annot=True,
            cmap="coolwarm")

plt.title("Correlation Heatmap")
plt.show()

X = clean_data.drop("Churn", axis=1)
y = clean_data["Churn"]

X = pd.get_dummies(X, drop_first=True)


#Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

#Scaling
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

#Model Definition
logistic_tuned = GridSearchCV(
    LogisticRegression(max_iter=2000),
    param_grid={"C": [0.01, 0.1, 1, 10, 50]},
    cv=5,
    scoring="roc_auc"
)
```

```
models = {
    "Logistic Regression (Tuned)": logistic_tuned,
    "Decision Tree": DecisionTreeClassifier(
        max_depth=5,
        random_state=42
    ),
    "Random Forest": RandomForestClassifier(
        n_estimators=300,
        random_state=42
    ),
    "Gradient Boosting": GradientBoostingClassifier(
        n_estimators=300,
        learning_rate=0.1,
        random_state=42
    ),
    "Extra Trees": ExtraTreesClassifier(
        n_estimators=300,
        random_state=42
    )
}
#Training
results = []

for name, model in models.items():

    if name == "Logistic Regression (Tuned)":
        cv_score = cross_val_score(model, X_train_scaled, y_train, cv=5, scoring='roc_auc').mean()
        model.fit(X_train_scaled, y_train)
        y_prob = model.predict_proba(X_test_scaled)[:, 1]
    else:
        cv_score = cross_val_score(model, X_train, y_train, cv=5, scoring='roc_auc').mean()
        model.fit(X_train, y_train)
        y_prob = model.predict_proba(X_test)[:, 1]

    y_pred = (y_prob > 0.5).astype(int)

    results.append({
        "Model": name,
        "CV ROC-AUC": cv_score,
        "Test Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred),
        "Recall": recall_score(y_test, y_pred),
        "F1 Score": f1_score(y_test, y_pred),
        "ROC-AUC": roc_auc_score(y_test, y_prob)
    })

results_df = pd.DataFrame(results)
results_df = results_df.sort_values(by="Test Accuracy", ascending=False)
```

```
results_df

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=["No Churn", "Churn"],
            yticklabels=["No Churn", "Churn"])

plt.title("Confusion Matrix")
plt.show()
if hasattr(best_model, "feature_importances_"):
    importances = pd.Series(best_model.feature_importances_, index=X.columns)
    importances.sort_values(ascending=False).head(10).plot(kind='bar')
    plt.title("Top 10 Important Features")
    plt.show()

elif best_model_name == "Logistic Regression (Tuned)":
    best_estimator = best_model.best_estimator_
    coef = pd.Series(best_estimator.coef_[0], index=X.columns)
    coef.sort_values(ascending=False).head(10).plot(kind='bar')
    plt.title("Top 10 Important Features (Logistic Coefficients)")
    plt.show()

else:
    print("Feature importance not available for this model.")
metrics = ["Test Accuracy", "Precision", "Recall", "F1 Score", "ROC-AUC"]

results_df.set_index("Model")[metrics].plot(
    kind="bar",
    figsize=(12,6)
)
plt.title("Performance Comparison of ML Models")
plt.ylabel("Score")
plt.xticks(rotation=30)
plt.legend(loc="lower right")
plt.show()

from sklearn.metrics import roc_curve
fpr, tpr, _ = roc_curve(y_test, y_prob)
plt.plot(fpr, tpr, label=best_model_name)
plt.plot([0,1], [0,1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
best_model_name = results_df.iloc[0]["Model"]
print("Best Model is:", best_model_name)
print("Accuracy:", round(results_df.iloc[0]["Test Accuracy"]*100, 3), "%")
```

CHAPTER 3

Outcomes

RESULT & OUTPUT

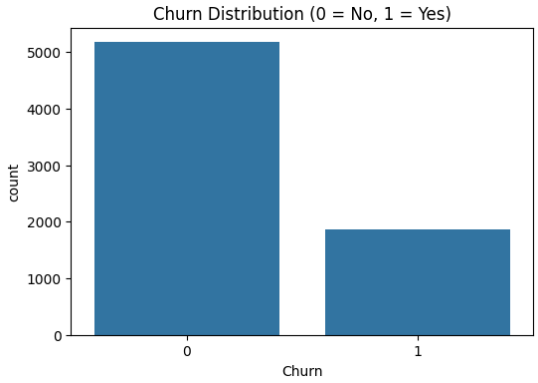


Fig 1: Churn Distribution (Bar Chart)

A count plot showing the frequency of "Yes" vs. "No" in the target variable to visualize class imbalance.

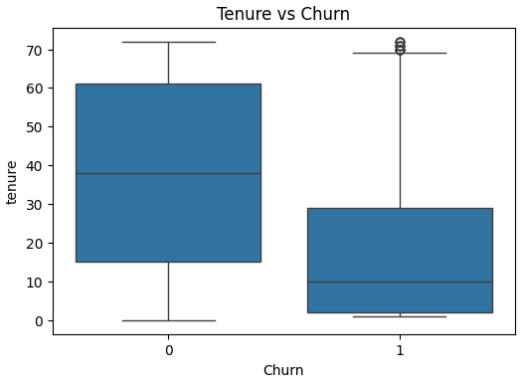


Fig 2: Tenure vs. Churn (Boxplot)

A plot showing that churned customers generally have lower median tenure compared to loyal customers.

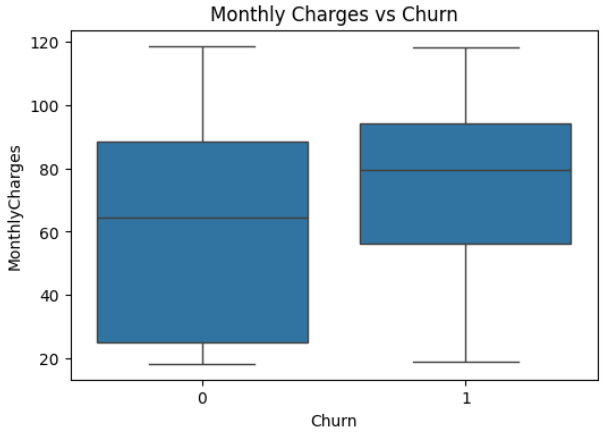


Fig 3: Monthly Charges vs. Churn (Boxplot)

A visual showing that customers who churn often have higher monthly costs.

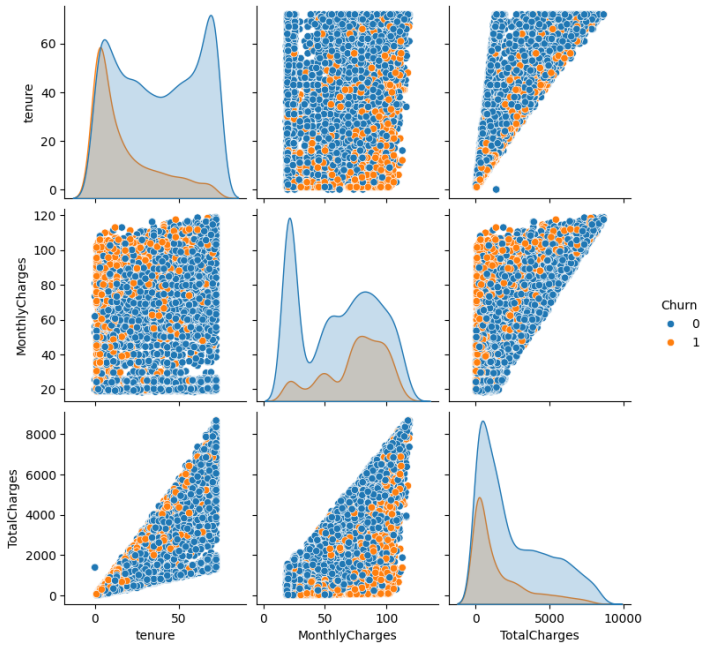


Fig 4: Feature Correlation Matrix

A pair plot visualizing the relationships between Tenure, MonthlyCharges, and TotalCharges, showing that churn (orange) is most frequent among new customers with high monthly costs.

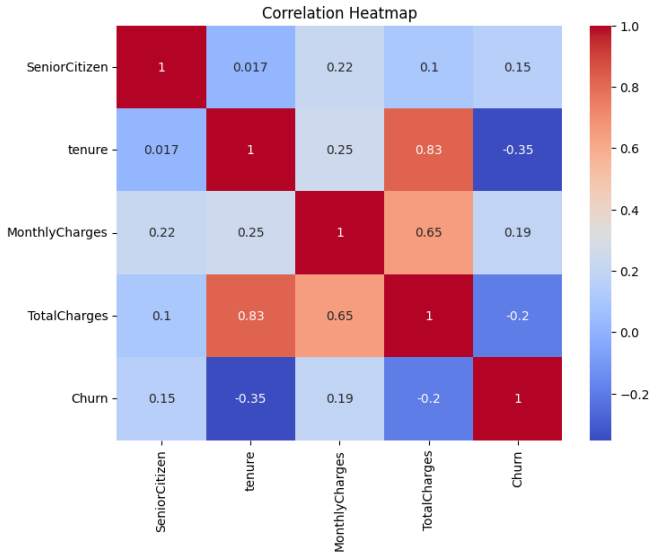


Fig 5: Feature Correlation Heatmap

A graphical matrix representing the Pearson correlation coefficients between numerical variables, used to identify multi-collinearity and key predictors.

Model	CV ROC-AUC	Test Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression (Tuned)	0.845921	0.801278	0.647799	0.550802	0.595376	0.840835
Decision Tree	0.817860	0.794180	0.629630	0.545455	0.584527	0.828358
Random Forest	0.825210	0.789922	0.631757	0.500000	0.558209	0.826969
Gradient Boosting	0.838074	0.787793	0.624585	0.502674	0.557037	0.832779
Extra Trees	0.800723	0.775727	0.596026	0.481283	0.532544	0.800801

Fig 6: Algorithm Benchmarking

A grouped bar chart comparing the performance of Logistic Regression, SVM, KNN, and Random Forest across Accuracy, Precision, and Recall.

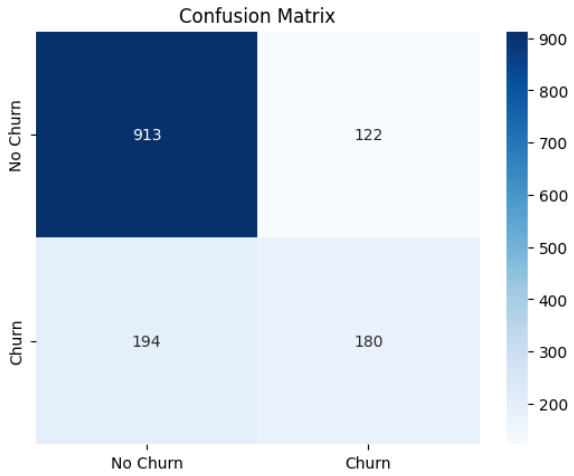


Fig 7: Confusion Matrix (Optimal Model)

A visualization of the Random Forest performance, showing the True Positives (correctly identified churners) and False Positives.

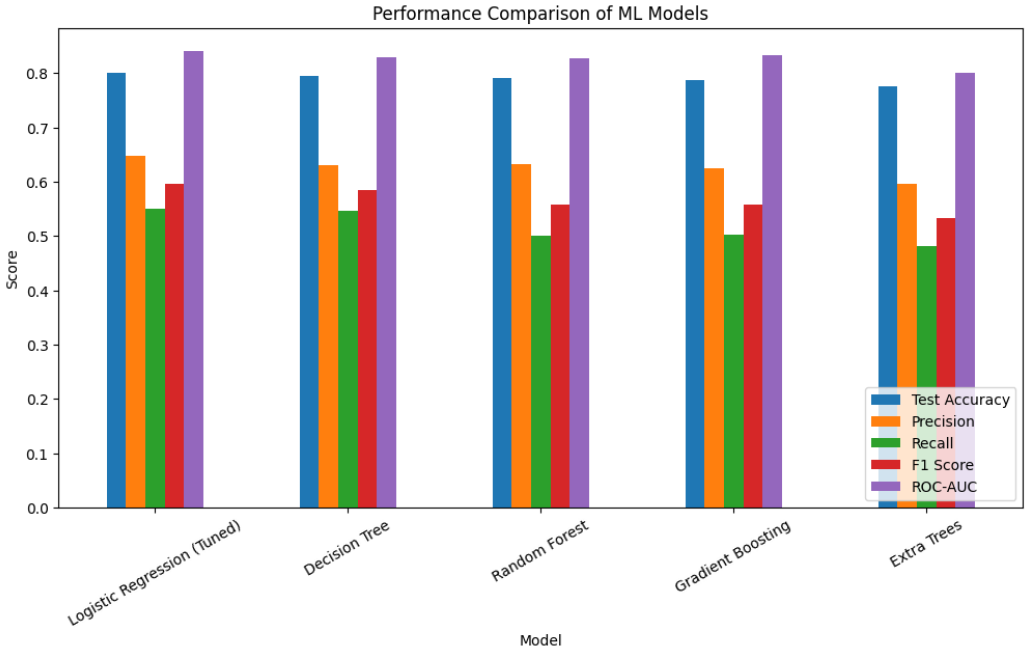


Fig 8: Comparison Chart

Bar chart comparing five ML models across Accuracy, Precision, Recall, F1, and ROC-AUC.

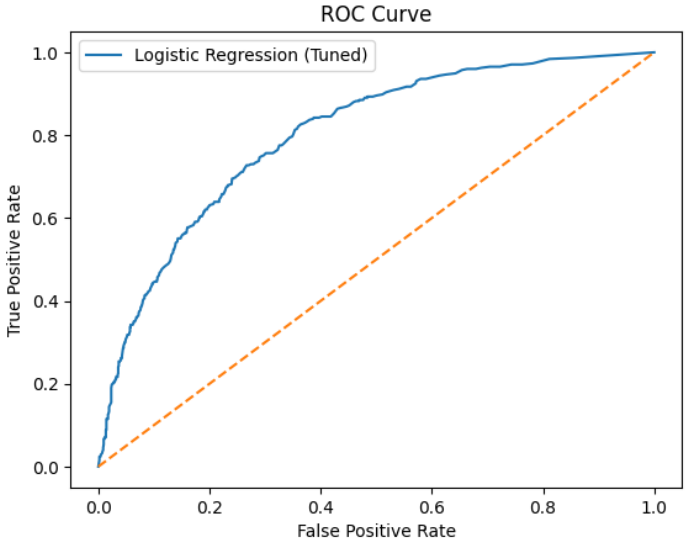


Fig 9: ROC Curve & AUC

A graph plotting the True Positive Rate against the False Positive Rate, with the Area Under the Curve (AUC) used to measure the model's discriminative power.

CONCLUSIONS

1. The main aim of this project was to build a machine learning system that could forecast the churn of customers based on demographic, account, and service-related variables. The machine learning system developed in this project was able to successfully combine data preprocessing, exploratory data analysis, feature engineering, model training, hyperparameter tuning, and evaluation of the model into a single process. The outcome of this project shows that machine learning can be used to accurately detect customers who are likely to leave a telecom service provider.

2. Data preprocessing was an important aspect that contributed to the enhancement of the model's reliability. The method used to handle missing values in the TotalCharges column, which involved the use of median imputation, ensured that the data was consistent and that the distribution was not affected. The process of removing irrelevant features like customerID ensured that the model was not noisy. The process of encoding categorical variables and scaling features ensured that the algorithms handled the variables efficiently.

3. Exploratory Data Analysis helped to identify important behavioral trends linked to churn. Customers with lower tenure were found to have a higher probability of churn, indicating that early-stage customers are more vulnerable. Higher monthly fees also had a moderate positive correlation with churn. The correlation heatmap validated tenure as the most negatively correlated predictor with churn, thus reiterating its significance in modeling.

4. Various classification models like Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Extra Trees were considered for comparison. Ensemble models like Random Forest and Gradient Boosting performed better in terms of overall accuracy and ROC-AUC scores, thus proving their efficacy in handling non-linear relationships in telecom data. Logistic Regression, when optimized, performed well in ROC-AUC scores and had a higher recall value, thus making it suitable when the sensitivity of churn detection is of prime importance.

5. Evaluation of the models using other metrics rather than accuracy was very important. Precision, Recall, F1-score, and ROC-AUC were used to get a better insight into the classification accuracy, especially with the moderate level of class imbalance. The confusion matrix helped in understanding the need to balance the detection of churners with the need to avoid false alarms. The ROC curve further helped in ascertaining that the chosen models were very good discriminators, with AUC values well above the random classification line.

From the above analysis, it is clear that machine learning-based churn prediction models can be an effective decision-support system for telecom firms.

SCOPE FOR FUTURE WORK

In the future, this system can be converted into a real-time customer monitoring tool. Instead of analyzing data only once, the model can be connected to the company's database so that it continuously checks customer activity and updates churn risk regularly. This would allow the company to identify at-risk customers immediately and take action before they leave.

The system can be integrated directly into the company's customer relationship management (CRM) software. When a customer's churn risk becomes high, the system can automatically notify the marketing or customer support team. This would make the solution more practical and directly usable in business operations rather than being only a predictive analysis model.

The prediction results can be linked to automated retention strategies. For example, if a customer is identified as high-risk, the system can automatically generate a discount offer, loyalty points, or a personalized message. This would transform the model from a prediction tool into a decision-making support system.

More customer information can be included to improve accuracy. In future work, additional data such as customer complaints, service quality issues, call drop frequency, internet speed records, or payment delays can be added. More detailed information will help the system understand customer dissatisfaction better.

The project can be expanded to predict not only whether a customer will leave, but also when they are likely to leave. This would help the company prioritize customers who are likely to churn soon and take urgent preventive measures.

The model can be tested on real-world datasets from different telecom companies or even other industries such as banking, insurance, or subscription-based services. This would check whether the approach works well across different types of businesses.

A simple dashboard or web application can be developed to make the system user-friendly. Managers who do not have technical knowledge should be able to upload customer data and view churn risk results through graphs and reports. This would make the system accessible to non-technical users.

Finally, the system can be improved by studying customer feedback after retention offers are given. This would help the company understand which retention strategies are most effective and continuously improve customer satisfaction.

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